

PORTFOLIO PROJECT

# Bike Share Rider Analysis

How do casual riders and annual members use bikes differently?

**10,000+**

Bikes Analyzed

**1,000+**

Stations

**2023**

Full Year Data

# The Business Problem

## The Goal

Cyclistic wants to convert casual riders into annual members — because annual memberships are far more profitable.

## Our job as analysts:

Find the behavioral differences between the two groups — and turn those findings into a targeted marketing strategy.

## Casual Riders

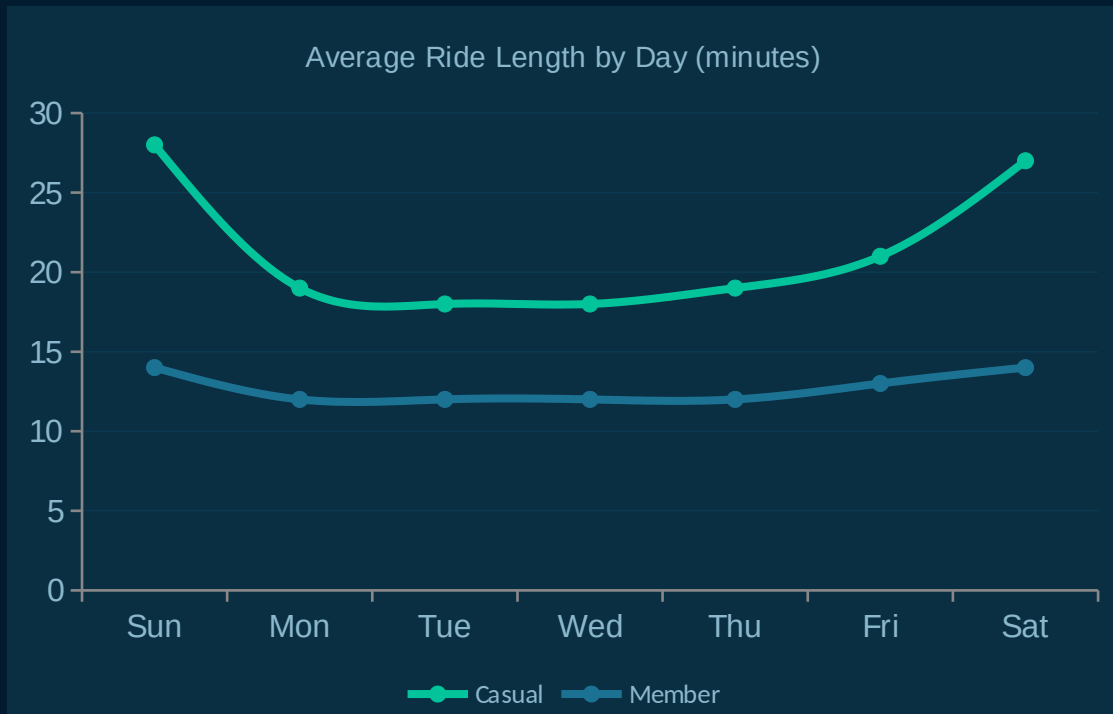
Fewer, longer rides — mostly weekends and leisure

## Annual Members

Frequent, shorter rides — consistent throughout the week

## USAGE PATTERNS

# How Each Group Rides



2×

Casual riders average twice the ride duration of members

## Weekend

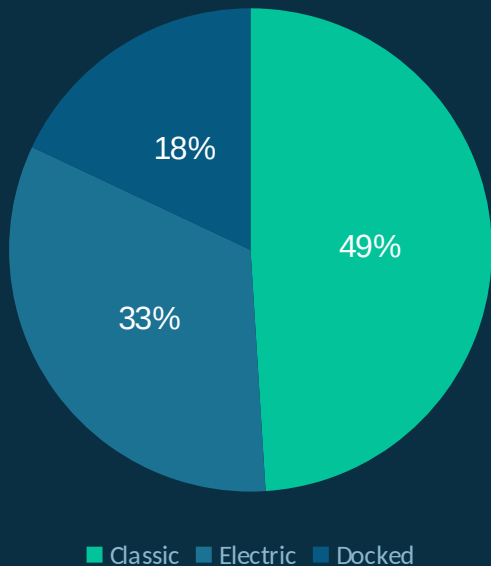
Casual rides peak on weekends; member rides stay flat all week

## Summer

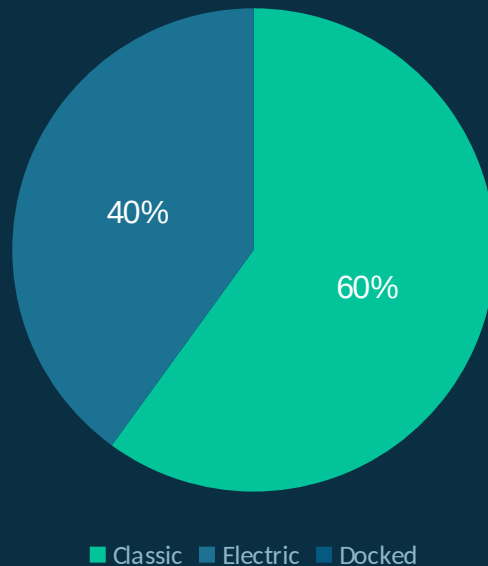
Both groups peak in summer — June through August

# What Each Group Rides

Casual Riders



Annual Members



Key insight: Only casual riders use docked bikes — members never do. Members use electric bikes twice as often as casual riders.

# Converting Casual Riders to Members

01

## Weekend Membership Promotions

Casual riders peak on weekends. Target them with trial membership offers on Saturday and Sunday — specifically at docking stations near parks and tourist areas.

02

## Seasonal Campaigns in Summer

Both groups ride most in summer. Launch annual membership deals in May–June before peak season hits, when casual riders are most active and motivated.

03

## Highlight Cost Savings

Show casual riders how many rides they take per season versus the cost of a membership. For frequent casual users, annual membership often pays for itself in 8–10 rides.

## TOOLS USED

- ▶ Python (Pandas, DuckDB, Matplotlib, Seaborn)
- ▶ Jupyter Notebook
- ▶ Divvy Tripdata (2023 Public Dataset)
- ▶ SQL queries via DuckDB for large dataset handling

**Have a dataset and a business question? Let's talk.**

One-time delivery · No maintenance required · Results in plain English